Design and Optimisation of an Automotive Rear Window Defroster

MULTIPHYSICS MODELLING USING COMSOL

ME46095

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Chapter 1

Model Definition

In this project, the design and optimisation of a resistive rear window defroster for an automotive application is investigated using finite element multiphysics modelling in COMSOL Multiphysics 6.2. The defroster must raise the entire glass surface temperature above the dew point to eliminate condensation and frost, while remaining within safe thermal and mechanical stress limits.

1.1 Problem Description

Rear window defrosters operate by passing electric current through thin conductive traces deposited on the inner surface of the glass. Joule heating in the traces conducts into the glass, raising its temperature above the dew point. The key design challenge is to achieve sufficiently uniform heating across the entire window using the minimum number of traces at a voltage compatible with a standard $12\,\mathrm{V}$ automotive electrical system.
The design must satisfy three main criteria:

1. **Thermal:** The minimum surface temperature of the glass must exceed the dew point temperature ($T_{\text{dew}} = 283.15\,\mathrm{K}$, i.e. $10\,^{\circ}\mathrm{C}$).
2. **Structural:** Thermal stresses from differential expansion between the trace material and glass must remain below material yield strengths.
3. **Transient:** The window must defrost within a reasonable time (target: $\leq 15\,\mathrm{min}$).

1.2 Geometry

The model is a two-dimensional representation of a flat rear window. The geometry consists of three components:

- A rectangular glass pane of dimensions $L_{\text{glass}} \times H_{\text{glass}} = 1.5\,\mathrm{m} \times 0.8\,\mathrm{m}$
- Horizontal conductive heating traces spanning the full width of the glass, created using a linear array
- Two vertical bus bars (left and right edges) connecting all traces to the power supply

The number of traces (n_{traces}) is a key design variable, swept from 8 to 22 in the parametric studies. The traces are uniformly spaced across the glass height. Figure 1.1 shows the COMSOL geometry for the baseline configuration.

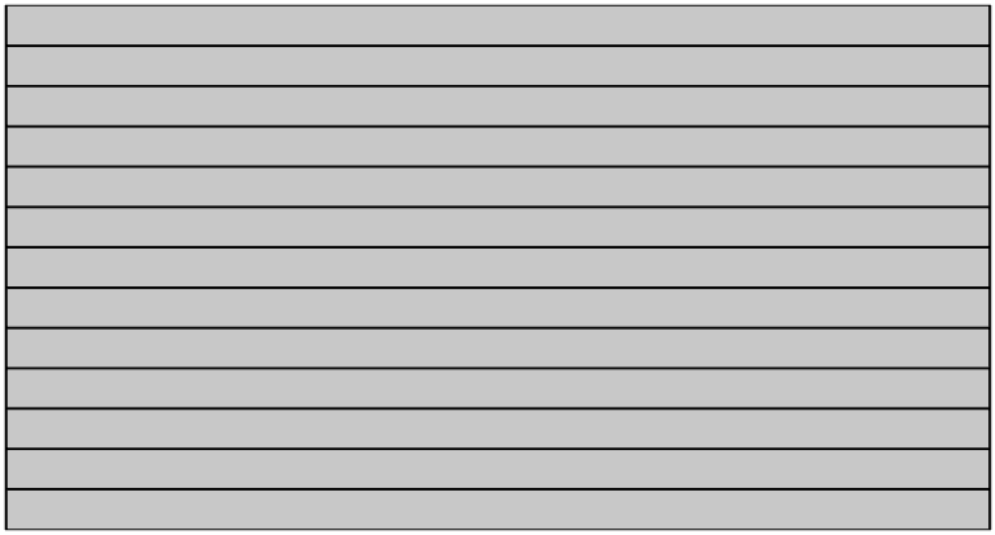


Figure 1.1: Two-dimensional geometry of the rear window defroster model showing the glass pane, horizontal heating traces, and vertical bus bars. Configuration shown: $n_{\text{traces}} = 12$.

The geometric parameters used in COMSOL are listed in Table 1.1.

Table 1.1: Geometric parameters of the rear window defroster model.

Parameter	**Symbol**	**Value**	**Description**
Glass length	L_{glass}	$1.5\,\text{m}$	Width of the window
Glass height	H_{glass}	$0.8\,\text{m}$	Height of the window
Glass thickness	t_{glass}	$5\,\text{mm}$	Out-of-plane thickness
Trace width	w_{trace}	$1.7\,\text{mm}$	Width of each heating trace
Trace thickness	t_{trace}	$0.11\,\text{mm}$	Thickness of each trace
Number of traces	n_{traces}	8–22 (swept)	Design variable

Since the model is two-dimensional with out-of-plane thickness, an effective 2D electrical conductivity is used for the traces:

$$\sigma_{\text{eff}} = \sigma_{\text{trace}} \cdot \frac{t_{\text{trace}}}{t_{\text{glass}}} \tag{1.1}$$

This accounts for the fact that the thin conductive traces occupy only a fraction of the out-of-plane glass thickness.

1.3 Materials

The window glass is modelled as tempered soda-lime glass using COMSOL's built-in material library. The key properties are listed in Table 1.2.

Table 1.2: Material properties of tempered soda-lime glass.

Property	**Value**	**Source**
Thermal conductivity (k)	$1.4\,\text{W}\,\text{m}^{-1}\,\text{K}^{-1}$	COMSOL library
Density (ρ)	$2500\,\text{kg}\,\text{m}^{-3}$	COMSOL library
Heat capacity (C_p)	$750\,\text{J}\,\text{kg}^{-1}\,\text{K}^{-1}$	COMSOL library
Fracture strength	$120\,\text{MPa}$	Literature

Four candidate trace materials were evaluated, with properties sourced from the CES EduPack database [1]. The full property set is given in Table 1.3.

Table 1.3: Trace material properties from CES EduPack [1]. σ: electrical conductivity, α: coefficient of thermal expansion, E: Young's modulus, ν: Poisson's ratio.

Material	$\boldsymbol{\sigma}$ **(**$\mathrm{S\,m^{-1}}$**)**	$\boldsymbol{\alpha}$ **(**$\mathrm{K^{-1}}$**)**	$\boldsymbol{E}$ **(**GPa**)**	$\boldsymbol{\nu}$	**Yield (**MPa**)**
Pd–Ag alloy	2.3×10^6	15×10^{-6}	102.5	0.38	410
Silver (comm. purity)	5.75×10^7	19.7×10^{-6}	71	0.39	190
Gold (soft annealed)	4.2×10^7	14×10^{-6}	78.5	0.42	20
Nichrome V	9.3×10^5	13×10^{-6}	212.5	0.29	365

1.4 Boundary Conditions

The boundary conditions represent a cold winter morning scenario where the exterior of the glass is exposed to ambient air and the interior faces the vehicle cabin. The values are listed in Table 1.4.

Table 1.4: Boundary conditions and environmental parameters.

Parameter	**Symbol**	**Value**	**Description**
Ambient temperature	T_{amb}	$275.15\,\mathrm{K}$ ($2\,^\circ\mathrm{C}$)	Exterior air
Cabin temperature	T_{cabin}	$293.15\,\mathrm{K}$ ($20\,^\circ\mathrm{C}$)	Interior cabin air
Initial temperature	T_{init}	$278.15\,\mathrm{K}$ ($5\,^\circ\mathrm{C}$)	Frosted window (transient)
Dew point temperature	T_{dew}	$283.15\,\mathrm{K}$ ($10\,^\circ\mathrm{C}$)	Target minimum temperature
External heat transfer coeff.	h_{ext}	$15\,\mathrm{W\,m^{-2}\,K^{-1}}$	Exterior convection
Internal heat transfer coeff.	h_{int}	$7\,\mathrm{W\,m^{-2}\,K^{-1}}$	Interior convection
Applied voltage	V_{in}	10–$14.5\,\mathrm{V}$ (swept)	Automotive 12V system range

Electrical. An electric potential V_{in} is applied on the left bus bar, and ground ($V = 0$) on the right bus bar. The electric current physics is defined only on the trace and bus bar domains (not the glass, which is electrically insulating).
Thermal. Since the model is 2D with out-of-plane thickness, heat losses to the exterior and interior are modelled using a volumetric heat source that accounts for convection on both sides of the glass:

$$Q_{\text{conv}} = \frac{h_{\text{ext}}(T_{\text{amb}} - T) + h_{\text{int}}(T_{\text{cabin}} - T)}{t_{\text{glass}}} \tag{1.2}$$

This expression replaces the out-of-plane heat flux boundary condition, which caused spatial frame conflicts in COMSOL 6.2 when coupled with solid mechanics.
Mechanical. For the stress analysis (Sim 2c), fixed constraints are applied on the left and right outer edges of the glass, representing the car body frame holding the window. This is a conservative simplification; in reality, a rubber seal provides some compliance.

1.5 Physics Coupling and Governing Equations

The model involves three coupled physics: Electric Currents, Heat Transfer in Solids, and Solid Mechanics. Their coupling is illustrated in Figure 1.2.

TODO: Create flow diagram — EC → Joule Heating → HT → Thermal Expansion → SM (draw manually)

Figure 1.2: Physics coupling in the multiphysics model. Joule heating from Electric Currents drives Heat Transfer, which in turn drives thermal expansion in Solid Mechanics.

Electric Currents. The electric field in the conductive traces is governed by current conservation:

$$\nabla \cdot \mathbf{J} = 0 \tag{1.3}$$

where $\mathbf{J} = \sigma_{\text{eff}}\mathbf{E}$ is the current density and $\mathbf{E} = -\nabla V$ is the electric field.
Heat Transfer in Solids. The heat equation in the glass domain is:

$$\rho C_p \frac{\partial T}{\partial t} = \nabla \cdot (k\nabla T) + Q_e + Q_{\text{conv}} \tag{1.4}$$

where $Q_e = \mathbf{J} \cdot \mathbf{E}$ is the resistive (Joule) heating from the Electromagnetic Heating multiphysics coupling, and Q_{conv} is the convective heat source from Equation (1.2).
For stationary studies, the left-hand side is zero and the equation reduces to:

$$\nabla \cdot (k\nabla T) + Q_e + Q_{\text{conv}} = 0 \tag{1.5}$$

Solid Mechanics. The mechanical equilibrium is described by:

$$\nabla \cdot \mathbf{S} + \mathbf{F}_v = 0 \tag{1.6}$$

where $\mathbf{S}$ is the Cauchy stress tensor. Thermal expansion introduces an initial strain:

$$\varepsilon_{\text{th}} = \alpha(T - T_{\text{ref}})\mathbf{I} \tag{1.7}$$

with reference temperature $T_{\text{ref}} = T_{\text{amb}} = 275.15\,\mathrm{K}$.

Chapter 2

Model Implementation in COMSOL

The design was developed iteratively through five simulations of increasing complexity. Each simulation builds on the previous one, progressively refining the design from a baseline model to a fully optimised transient solution. Table 2.1 summarises all simulations.

Table 2.1: Overview of simulations performed.

Simulation	Study Type	Sweep Parameters	Objective
Sim 1	Stationary	None (baseline)	Verify model physics
Sim 2	Stationary	V_{in}, n_{traces}	Identify viable operating range
Sim 2b	Stationary	Material, n_{traces}, V_{in}	Material optimisation
Sim 2c	Stationary (sequential)	Material, n_{traces}	Stress analysis
Sim 3	Time dependent	Material, n_{traces}, V_{in}	Defrost time
Sim 4	Time dependent	Zone layout (n_{top}, n_{mid}, n_{bot})	Zoned trace spacing
Sim 5	Time dependent	Chebyshev exponent p	Non-uniform spacing optimisation
Sim 6	Time dependent	Weather scenario	Robustness across climate conditions

2.1 Simulation 1 — Baseline Stationary Model

The first simulation establishes the baseline model with the default configuration: 12 traces of silver paste ($\sigma = 2\times10^6\,\mathrm{S\,m^{-1}}$) at $V_{\text{in}} = 12\,\mathrm{V}$. A stationary study couples Electric Currents and Heat Transfer via Electromagnetic Heating.
The purpose of Sim 1 is to verify that:

- The voltage gradient across the traces is correct (12 V at the left bus bar, 0 V at the right)
- The temperature distribution is physically realistic (in the range 270–320 K)
- The cold spots are located at the expected positions (top and bottom edges farthest from traces)

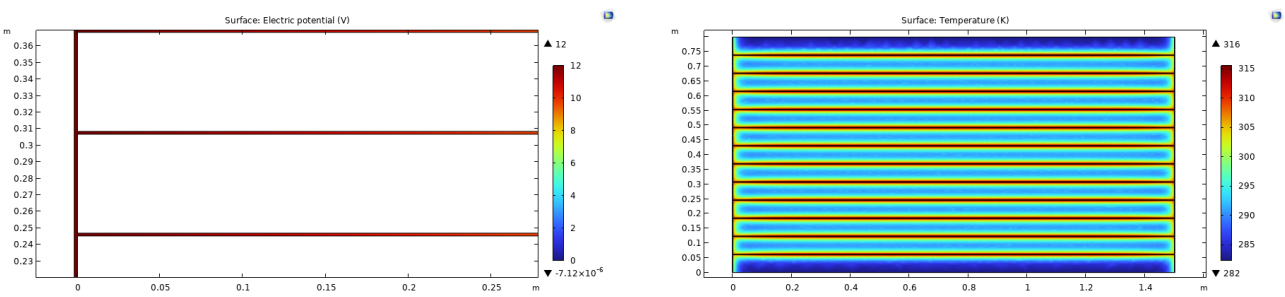


Figure 2.1: Sim 1 baseline results at $V_{\text{in}} = 12\,\mathrm{V}$, $n_{\text{traces}} = 12$, silver paste. (a) Electric potential distribution on the traces and bus bars. (b) Temperature distribution across the glass surface.

The baseline results show a surface-average temperature of $297.13\,\mathrm{K}$ (above the dew point) but a surface-minimum temperature of $282.12\,\mathrm{K}$ (below the dew point). The cold spots are at the top and bottom edges of the glass where trace coverage does not extend. This motivates the parametric sweeps in the following simulations.

2.2 Simulation 2 — Voltage and Trace Count Sweeps

Sim 2 investigates the sensitivity of the minimum glass temperature to the applied voltage and number of traces, independently and in combination.
A parametric sweep of V_{in} from $8\,\text{V}$ to $16\,\text{V}$ (9 values) is performed at $n_{\text{traces}} = 12$. Even at $16\,\text{V}$, the minimum temperature ($283.09\,\text{K}$) barely reaches the dew point, indicating that voltage alone cannot solve the problem with 12 traces.
A parametric sweep of n_{traces} from 8 to 22 is performed at $V_{\text{in}} = 12\,\text{V}$. Results are shown in Figure 2.2.

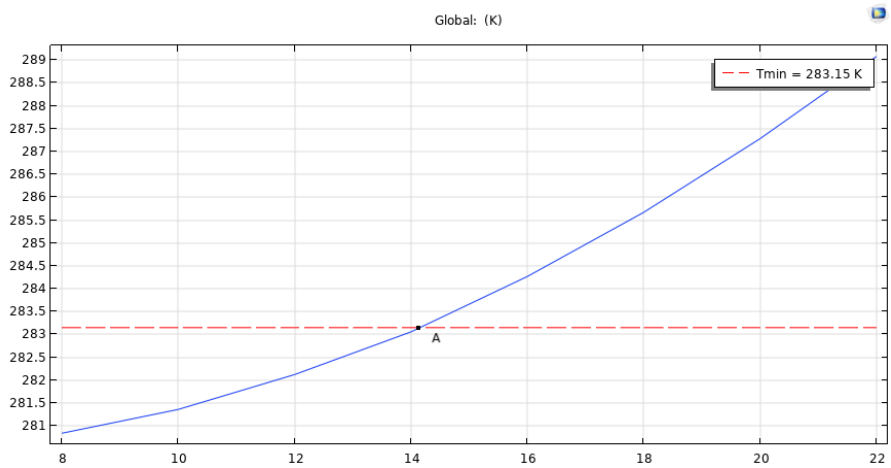


Figure 2.2: Minimum glass surface temperature as a function of the number of traces at $V_{\text{in}} = 12\,\text{V}$. The horizontal dashed line indicates the dew point temperature ($283.15\,\text{K}$).

A full factorial sweep over $n_{\text{traces}} \in \{8, 10, 12, 14, 16, 18, 20, 22\}$ and $V_{\text{in}} \in \{8, 10, 12, 14, 16\}$ V produces 35 stationary solutions. Results are shown in Table 2.2.

Table 2.2: Minimum surface temperature T_{min} (K) for the combined voltage–trace count sweep (silver paste, $\sigma = 2 \times 10^6\,\text{S}\,\text{m}^{-1}$). Values $\geq 283.15\,\text{K}$ are shown in **bold**.

n_{traces}	$8\,\text{V}$	$10\,\text{V}$	$12\,\text{V}$	$14\,\text{V}$	$16\,\text{V}$
8	280.86	280.85	280.84	280.83	280.81
10	281.09	281.21	281.36	281.54	281.74
12	281.43	281.74	282.12	282.57	**283.09**
14	281.85	282.39	**283.06**	**283.84**	**284.75**
16	282.38	**283.23**	**284.27**	**285.49**	**286.91**
18	**283.01**	**284.21**	**285.67**	**287.40**	**289.40**
20	**283.72**	**285.33**	**287.28**	**289.60**	**292.27**
22	**284.52**	**286.57**	**289.08**	**292.04**	**295.46**

The key conclusion is that **increasing the number of traces is more effective than increasing voltage**. The cheapest solution above the dew point is 16 traces at $10\,\text{V}$ ($283.23\,\text{K}$).

2.3 Simulation 2b — Material Optimisation

Sim 2b extends the parametric sweep to include four candidate trace materials from the CES EduPack database [1]. The sweep covers $4 \times 8 \times 4 = 128$ combinations (4 materials, 8 trace counts, 4 voltages).
The voltages used reflect the realistic automotive range: $10\,\text{V}$ (battery low), $12\,\text{V}$ (nominal), $13.5\,\text{V}$ (engine running), and $14.5\,\text{V}$ (maximum alternator output).
The minimum number of traces needed to reach the dew point for each material and voltage is shown in Table 2.3.

Table 2.3: Minimum number of traces to reach $T_{\text{min}} \geq 283.15\,\text{K}$ for each material and voltage.

Material	$10\,\text{V}$	$12\,\text{V}$	$13.5\,\text{V}$	$14.5\,\text{V}$
Pd–Ag alloy	16	14	14	14
Nichrome V	22	18	18	16
Silver (comm. purity)	*Overheats — not viable*			
Gold (soft annealed)	*Overheats — not viable*			

The full temperature results for the two viable materials (Pd–Ag alloy and Nichrome V) are shown in Tables 2.4 and 2.5.

Table 2.4: T_{min} (K) for Pd–Ag alloy ($\sigma = 2.3 \times 10^6\,\text{S}\,\text{m}^{-1}$). Values $\geq 283.15\,\text{K}$ shown in **bold**.

n_{traces}	$10\,\text{V}$	$12\,\text{V}$	$13.5\,\text{V}$	$14.5\,\text{V}$
8	280.85	280.83	280.82	280.82
10	281.26	281.43	281.58	281.69
12	281.87	282.31	282.69	282.97
14	282.62	**283.38**	**284.05**	**284.54**
16	**283.59**	**284.78**	**285.81**	**286.57**
18	**284.71**	**286.39**	**287.86**	**288.93**
20	**286.00**	**288.24**	**290.20**	**291.63**
22	**287.43**	**290.31**	**292.82**	**294.65**

Table 2.5: T_{min} (K) for Nichrome V ($\sigma = 9.3 \times 10^5\,\text{S}\,\text{m}^{-1}$). Values $\geq 283.15\,\text{K}$ shown in **bold**.

n_{traces}	$10\,\text{V}$	$12\,\text{V}$	$13.5\,\text{V}$	$14.5\,\text{V}$
8	280.87	280.86	280.86	280.85
10	281.03	281.10	281.16	281.21
12	281.28	281.46	281.61	281.72
14	281.58	281.89	282.16	282.36
16	281.97	282.45	282.87	**283.18**
18	282.43	**283.11**	**283.70**	**284.13**
20	282.95	**283.86**	**284.65**	**285.23**
22	**283.53**	**284.69**	**285.71**	**286.45**

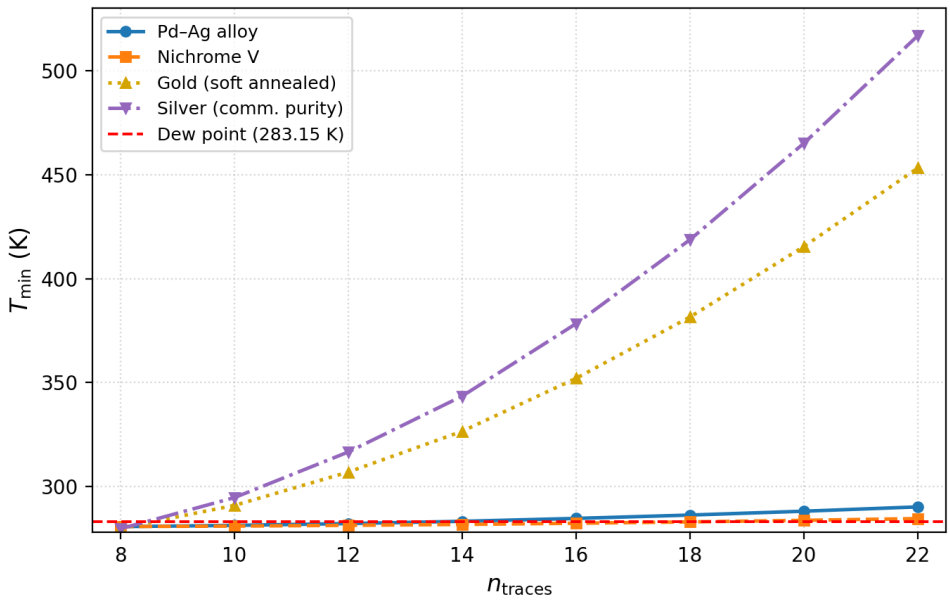


Figure 2.3: Comparison of minimum surface temperature vs. number of traces for all four EduPack materials at $V_{\text{in}} = 12\,\text{V}$. Silver and Gold exceed safe temperatures at $n_{\text{traces}} \geq 10$ and are not viable.

- **Silver (commercial purity) and Gold (soft annealed) are not viable** — their bulk conductivity ($> 10^7$ S/m) is too high for the screen-printed geometry, causing extreme overheating (Silver reaches $625\,\mathrm{K}$ at 22 traces, $14.5\,\mathrm{V}$).
- **Pd–Ag alloy is the best EduPack material** — it reaches the dew point at 14 traces ($12\,\mathrm{V}$+). Temperatures remain in a safe range.
- **Nichrome V is viable but requires more traces** (18–22 depending on voltage).

2.4 Simulation 2c — Stress Analysis

Sim 2c evaluates the thermal stresses for the two shortlisted materials (Pd–Ag alloy and Nichrome V). Solid Mechanics and Thermal Expansion are re-enabled, and a sequential solve is used:

- **Step 1:** Electric Currents + Heat Transfer (stationary)
- **Step 2:** Solid Mechanics with Thermal Expansion (stationary, using Step 1 temperatures)

This sequential approach avoids the spatial frame conflict ("dvol_spatial undefined") encountered when solving all physics simultaneously in COMSOL 6.2.

Table 2.6: Von Mises stress results at $V_{\text{in}} = 12\,\mathrm{V}$.

Material	n_{traces}	**Max stress (MPa)**	**Yield (MPa)**	**Safety factor**
Pd–Ag alloy	14	120.3	410	3.4×
Pd–Ag alloy	18	134.5	410	3.0×
Nichrome V	14	94.6	365	3.9×
Nichrome V	18	102.9	365	3.5×

TODO (Sebtain): Sim2c — von Mises stress, Pd–Ag, 14 traces

TODO (Sebtain): Sim2c — von Mises stress, Nichrome V, 18 traces

Figure 2.4: Von Mises stress distribution at $V_{\text{in}} = 12\,\mathrm{V}$. (a) Pd–Ag alloy, $n_{\text{traces}} = 14$. (b) Nichrome V, $n_{\text{traces}} = 18$. Peak stresses concentrate at trace endpoints near the fixed edges.

Both materials pass the stress constraint with safety factors $> 3\times$. Peak stresses ($\sim 120\,\mathrm{MPa}$) concentrate at trace endpoints near the fixed boundary — a conservative artifact of the fully fixed edge condition. The glass bulk stress remains very low ($2\,\mathrm{MPa}$ to $4\,\mathrm{MPa}$), well below the $120\,\mathrm{MPa}$ fracture strength of tempered glass.

2.5 Simulation 3 — Transient Defrost Study

Sim 3 evaluates how long each configuration takes to defrost the window from the initial frosted condition ($T_{\text{init}} = 278.15\,\mathrm{K}$) to the point where $T_{\text{min}} \geq T_{\text{dew}}$.
A time-dependent study is used with output times from 0 to $3600\,\mathrm{s}$ in $60\,\mathrm{s}$ increments. The Electric Currents equation is set to stationary form (the electric field equilibrates in microseconds compared to the thermal time scale of minutes). Solid Mechanics is disabled for computational efficiency.

The sweep covers: 2 materials $\times$ 5 trace counts (14, 16, 18, 20, 22) $\times$ 2 voltages (12, $14.5\,\mathrm{V}$) = 20 runs. The mesh was refined to Fine (189,192 domain elements) for accuracy.
Defrost times for all configurations are summarised in Table 2.7.

Table 2.7: Defrost time (time for T_{min} to reach $T_{\text{dew}} = 283.15\,\mathrm{K}$) for all Sim 3 configurations. "Never" indicates the steady-state T_{min} remains below the dew point at that voltage.

Material	n_{traces}	$12\,\mathrm{V}$	$14.5\,\mathrm{V}$
Pd–Ag alloy	14	41 min	22 min
Pd–Ag alloy	16	22 min	16 min
Pd–Ag alloy	18	16 min	12 min
Pd–Ag alloy	20	**12 min**	**10 min**
Pd–Ag alloy	22	10 min	8 min
Nichrome V	14	Never	Never
Nichrome V	16	Never	59 min
Nichrome V	18	Never	25 min
Nichrome V	20	28 min	18 min
Nichrome V	22	20 min	14 min

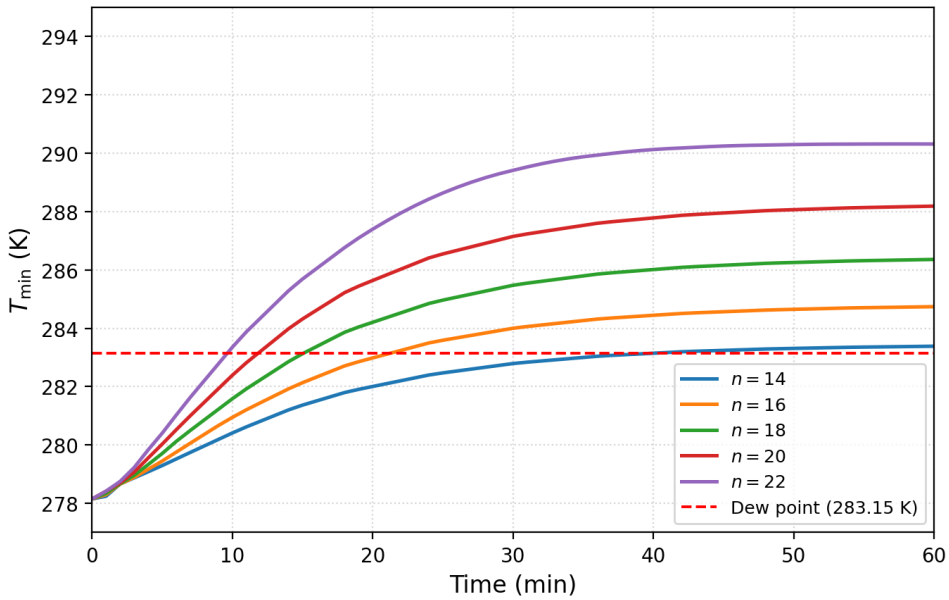


Figure 2.5: Transient evolution of minimum surface temperature for Pd–Ag alloy at $V_{\text{in}} = 12\,\mathrm{V}$ for different numbers of traces. The horizontal dashed line indicates the dew point.

Based on these results, **Pd–Ag alloy with $n_{\text{traces}} = 20$ at $V_{\text{in}} = 12\,\mathrm{V}$ is selected as the recommended design**. The key reasons are:

- **Battery-only operation:** A nominal automotive 12 V system operates directly from the battery without the engine running [2]. This is critical for a rear defroster, which must function during a cold start before the engine—and therefore the alternator—is active. Voltages of $13.5\,\mathrm{V}$ and $14.5\,\mathrm{V}$ are only available when the alternator is charging, requiring the engine to be running.
- **Acceptable defrost time:** At $12\,\mathrm{V}$, 20 traces defrosts the window in $12\,\mathrm{min}$ — well within a practical warm-up period.
- **Adequate thermal margin:** The steady-state $T_{\text{min}} = 288.2\,\mathrm{K}$ is $5.05\,\mathrm{K}$ above the dew point, providing a comfortable margin against condensation re-formation.
- **Diminishing returns at 22 traces:** Increasing to 22 traces reduces defrost time by only 2 minutes (10 vs. 12 min at $12\,\mathrm{V}$) at the cost of higher material use and increased trace density.

An alternative configuration of 22 traces at $14.5\,\mathrm{V}$ achieves defrost in $8\,\mathrm{min}$ but requires the engine to be running and is therefore only suitable as a secondary high-performance mode.

2.6 Simulation 4 — Trace Pattern Optimisation

After establishing the optimal material (Pd–Ag alloy) and operating conditions in the previous simulations, Sim 4 investigates the effect of trace layout on the temperature uniformity across the glass surface. The baseline simulations revealed that cold spots form at the top and bottom edges of the glass, farthest from any heating trace. A natural mitigation strategy is to concentrate more traces near these edges.
The glass is divided into three horizontal zones with independent trace densities:

- **Top zone:** $H_{\text{top}} = 200\,\text{mm}$, $n_{\text{top}} = 6$ traces
- **Middle zone:** $H_{\text{mid}} = 400\,\text{mm}$, $n_{\text{mid}} = 8$ traces
- **Bottom zone:** $H_{\text{bot}} = 200\,\text{mm}$, $n_{\text{bot}} = 6$ traces

This gives a total of 20 traces (same as the recommended uniform design) but with denser spacing in the edge zones. The edge gap is $28.6\,\text{mm}$ (vs. $38.1\,\text{mm}$ uniform), and the centre gap is $44.4\,\text{mm}$.
Zone parameters were added to Global Definitions and each zone was created as a separate Rectangle + Array in the geometry, with spacing $H_{\text{zone}}/(n_{\text{zone}} + 1)$. The old uniform-spacing geometry (Array 1 + Rectangle 2) was disabled.

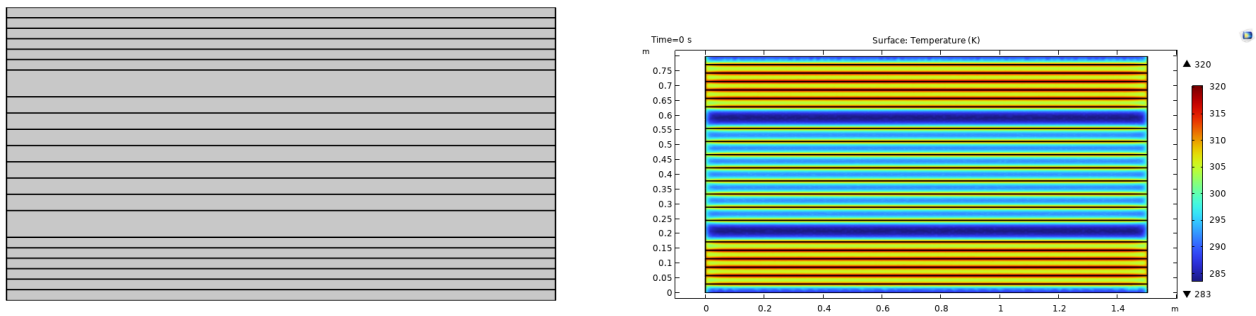


Figure 2.6: Zoned trace layout (Approach A). (a) Geometry with 6/8/6 distribution across top/middle/bottom zones. (b) Temperature distribution at $T = T_{\text{dew}}$ showing reduced cold spots at the edges.

The study was changed from stationary to time dependent with output range `range(0,60,3600)`, i.e. every $60\,\text{s}$ for $60\,\text{min}$. The Electric Currents equation form was set to Stationary (as in Sim 3) since the electric field equilibrates instantaneously compared to the thermal time scale. The initial temperature was set to $T_{\text{init}} = 278.15\,\text{K}$.
The zoned approach was not pursued further because the continuous Chebyshev spacing (Sim 5) proved far more effective, as discussed in Section 2.7.

2.7 Simulation 5 — Chebyshev Trace Spacing Optimisation

While the zoned approach (Sim 4) provides a discrete improvement, a mathematically elegant alternative is to use a continuous spacing function that naturally clusters traces near the edges. Chebyshev node spacing, derived from the roots of Chebyshev polynomials, is well known in numerical analysis for minimising interpolation error by placing points near interval boundaries. This property is analogous to the defroster design goal of concentrating heating where it is most needed.

2.7.1 Spacing Formula and Implementation

The vertical position y_i of the i-th trace ($i = 1, \ldots, 20$) is given by:

$$y_i = \frac{H_{\text{glass}}}{2}\left(1 - \text{sign}\left(\cos\frac{i\pi}{21}\right)\left|\cos\frac{i\pi}{21}\right|^p\right) \tag{2.1}$$

where p is a clustering exponent that controls how aggressively the traces are pushed toward the edges:

- $p < 1$: More uniform than standard Chebyshev (less edge clustering)
- $p = 1$: Standard Chebyshev distribution
- $p > 1$: More aggressive edge clustering

The sign$(\cdot)|\cdot|^p$ formulation preserves the symmetry of the cosine distribution for all values of p, including non-integer exponents.
Twenty individual rectangles were created in the geometry, each with a unique y-position computed from Equation (2.1). The exponent p was defined as a global parameter, enabling a parametric sweep. All other model settings (material: Pd–Ag alloy, σ_{eff} correction, time-dependent study, boundary conditions) were identical to Sim 4.

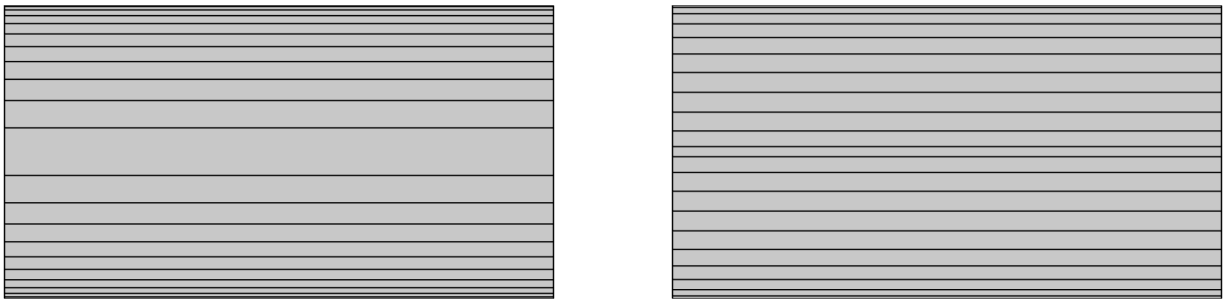


Figure 2.7: Chebyshev trace spacing for (a) $p = 0.7$ (near-uniform) and (b) $p = 1.3$ (optimal). Edge gaps decrease significantly with increasing p while centre gaps widen.

2.7.2 Results and Optimum

Three values of p were tested first to establish the feasibility of the approach:

- $p = 1.0$ (standard Chebyshev): **defrost in** $7\,\text{min}$, steady-state $T_{\text{min}} = 292.6\,\text{K}$
- $p = 0.85$: **never reaches dew point** — steady-state $T_{\text{min}} = 282.06\,\text{K}$
- $p = 0.7$: **never reaches dew point** — steady-state $T_{\text{min}} = 282.06\,\text{K}$

The $p = 1.0$ result is dramatic: defrost time is reduced from $12\,\text{min}$ (uniform) to $7\,\text{min}$ — a 42% improvement with exactly the same number of traces, same material, and same voltage. However, reducing p below 1.0 removes too much edge clustering and the cold spots return.
To find the optimal exponent, a parametric sweep was performed over $p \in \{0.7, 0.8, 0.9, 0.95, 1.0, 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1$
All runs use 20 traces of Pd–Ag alloy at $V_{\text{in}} = 12\,\text{V}$. The trace y-positions were expressed parametrically in terms of p, so COMSOL automatically updates the geometry for each sweep value.
The results are presented in Table 2.8 and plotted in Figure 2.8.

Table 2.8: Chebyshev exponent sweep results (Pd–Ag alloy, 20 traces, $V_{\text{in}} = 12\,\text{V}$). $T_{360\text{s}}$: minimum temperature at $t = 6\,\text{min}$.

p	T_{min} at $t = 360\,\text{s}$ (K)	**Defrost time**	**Steady-state** T_{min} (K)
0.70	279.53	Never	282.1
0.80	279.72	23 min	284.0
0.90	280.44	12 min	287.5
1.00	282.23	8 min	292.6
1.10	282.97	7 min	294.5
1.20	283.16	6 min	294.9
1.30	**283.20**	**6 min**	**295.0**
1.40	283.09	6 min	294.8
1.50	282.97	7 min	294.5
1.60	282.91	7 min	294.4
1.70	282.74	7 min	293.9
1.80	282.56	7 min	293.4
1.90	282.40	7 min	293.1
2.00	282.28	7 min	292.8
3.00	281.17	9 min	289.9

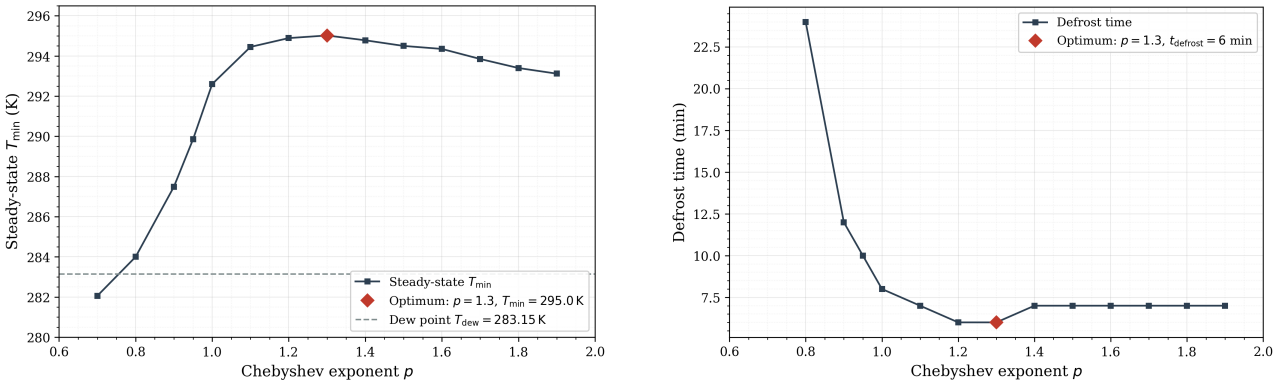


Figure 2.8: Effect of the Chebyshev exponent p on (a) steady-state minimum temperature and (b) defrost time. The optimum is $p = 1.3$, achieving the highest T_{min} and fastest defrost.

The optimum is clearly at $p = 1.3$: it achieves the highest minimum temperature at the 6-minute mark ($283.20\,\text{K}$) and the highest steady-state T_{min} ($295.0\,\text{K}$). The physical interpretation is:

- $p < 1.0$**:** Insufficient edge clustering. The edges remain cold, and the centre is over-heated with closely spaced traces — reproducing the original problem.
- $p = 1.0$**–**1.3**:** Increasing p progressively moves more traces to the edges, heating the cold spots more effectively. The centre gap widens but the glass thermal conductivity is sufficient to bridge it.
- $p > 1.3$**:** Over-clustering at edges. The centre becomes a cold spot because the traces are too far apart. At $p = 2.0$ the cosine-squared creates three clusters (two edges plus centre), and at $p = 3.0$ the clustering is too sharp for the window height.

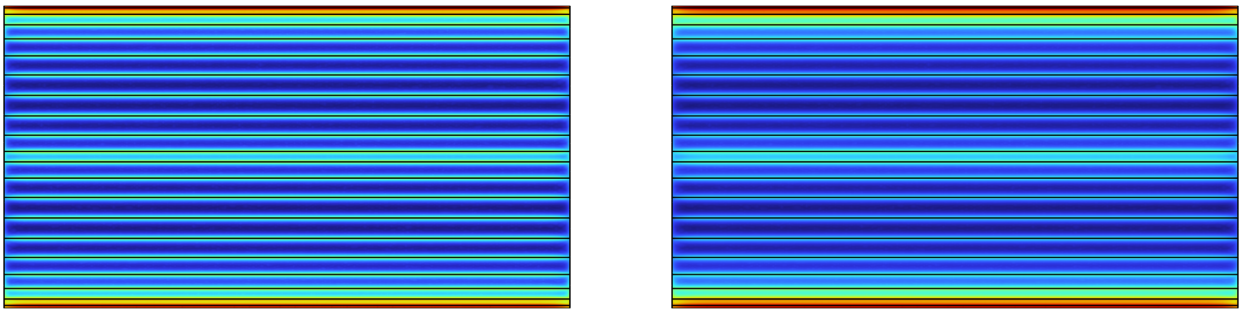


Figure 2.9: Temperature distribution for Chebyshev spacing $p = 1.3$ (Pd–Ag alloy, 20 traces, $12\,\text{V}$). (a) At $t = 6\,\text{min}$ (defrost moment): $T_{\text{min}} = 283.20\,\text{K}$. (b) Steady state: $T_{\text{min}} = 295.0\,\text{K}$.

2.7.3 Practical Considerations

An important practical constraint for rear window trace patterns is driver visibility. Per the SAE J216 standard, obscuration of the rear vision area should not exceed 5% in the primary zone [3]. For 20 traces with $w_{\text{trace}} = 1.7\,\text{mm}$:

$$\text{Obscuration} = \frac{n_{\text{traces}} \times w_{\text{trace}}}{H_{\text{glass}}} = \frac{20 \times 1.7}{800} = 4.25\% \tag{2.2}$$

This is well within the 5% limit regardless of the spacing pattern, since the total trace area is unchanged. The Chebyshev spacing only redistributes the same number of traces, so visibility is not affected.
Adding vertical traces between the horizontal ones was considered to create a grid pattern. However, analysis showed this would be ineffective: since all parallel horizontal traces are at the same voltage at any given x-position, vertical connectors between them would have zero potential difference ($\Delta V = 0$) and carry no current. Variable-width traces could create voltage differences enabling vertical current flow, but this adds significant manufacturing complexity without clear benefit over optimising the horizontal spacing alone.

2.8 Simulation 6 — Weather Condition Robustness Study

All previous simulations used a fixed ambient condition: $T_{\text{amb}} = 2\,°\text{C}$, $h_{\text{ext}} = 15\,\text{W}\,\text{m}^{-2}\,\text{K}^{-1}$, and $T_{\text{dew}} = 10\,°\text{C}$. Sim 6 tests the final design (Pd–Ag alloy, 20 traces, Chebyshev $p = 1.3$, $V_{\text{in}} = 12\,\text{V}$) across five realistic weather scenarios to verify that it performs robustly under a range of real-world conditions.

2.8.1 Weather Scenarios

The five scenarios were selected to cover the full range of conditions a vehicle might encounter in Northwestern Europe [4], from mild autumn fog to extreme winter frost. Each scenario defines a physically consistent set of T_{amb}, T_{dew}, and h_{ext} values — the dew point is not fixed but varies with ambient temperature and relative humidity [5] — as summarised in Table 2.9.

Table 2.9: Weather scenarios tested in Sim 6. Each scenario uses a physically realistic dew point consistent with the ambient humidity conditions.

#	Scenario	T_{amb} (°C)	T_{dew} (°C)	h_{ext} ($\text{W}\,\text{m}^{-2}\,\text{K}^{-1}$)
1	Mild winter	2	10	15
2	Cold winter	-5	0	25
3	Extreme cold	-20	-15	30
4	Cool autumn	10	13	10
5	Freezing rain	-2	0	40

2.8.2 Results

The transient T_{min} for each scenario is shown in Figure 2.10. Each solid curve is paired with a dashed horizontal line at its scenario-specific dew point. Defrost is achieved when the curve crosses its corresponding dashed line.

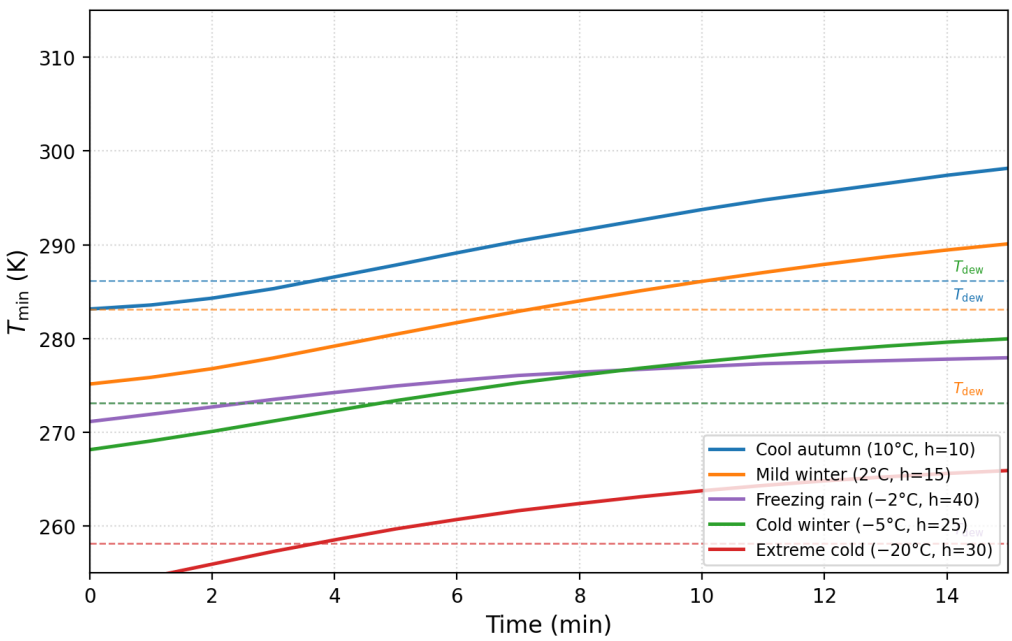


Figure 2.10: Transient minimum surface temperature for all five weather scenarios (Pd–Ag alloy, 20 traces, Chebyshev $p = 1.3$, $V_{\text{in}} = 12\,\text{V}$). Each dashed line shows the scenario-specific dew point. All curves cross their dew point within $8\,\text{min}$.

A summary of defrost times, steady-state minimum temperatures, and margins above the dew point is given in Table 2.10.

Table 2.10: Sim 6 results for all five weather scenarios. Margin $= T_{\text{min,SS}} - T_{\text{dew}}$.

#	Scenario	Defrost time	$T_{\text{min,SS}}$ (K)	T_{dew} (K)	Margin (K)
1	Mild winter	7 min	295.0	283.15	+11.9
2	Cold winter	4 min 47 s	281.6	273.15	+8.5
3	Extreme cold	3 min 41 s	267.0	258.15	+8.9
4	Cool autumn	3 min 40 s	307.5	286.15	+21.4
5	Freezing rain	2 min 33 s	278.6	273.15	+5.4

All five scenarios pass the defrost criterion with positive margin. Key observations are:

- **Faster defrost under harsher convection:** Scenarios with high h_{ext} (freezing rain: 40, extreme cold: 30) defrost in under 4 minutes. The high convective coefficient drives a steeper thermal gradient, which accelerates the initial transient even though the steady-state temperature is lower.
- **Tightest margin in freezing rain (Scenario 5):** With $h_{\text{ext}} = 40\,\mathrm{W\,m^{-2}\,K^{-1}}$ and a high convective heat loss, the steady-state T_{min} is only $5.4\,\mathrm{K}$ above the dew point. This is the worst-case scenario for sustained condensation prevention.
- **Largest margin in cool autumn (Scenario 4):** Low convective loss and mild ambient conditions result in a $21.4\,\mathrm{K}$ margin — more than sufficient.
- **Robustness across $\Delta T_{\text{amb}} = 30\,\mathrm{K}$:** The design passes all scenarios spanning a $30\,\mathrm{K}$ ambient temperature range ($-20\,^\circ\mathrm{C}$ to $10\,^\circ\mathrm{C}$), demonstrating robustness to real-world variability.

Chapter 3

Model Validation

3.1 Mesh Convergence Study

A mesh convergence study was performed to verify that all results are independent of mesh resolution. The study was carried out using a stationary Electric Currents and Heat Transfer solve under the Sim 6 Scenario 1 conditions (mild winter: $T_{\text{amb}} = 2\,^{\circ}\mathrm{C}$, $h_{\text{ext}} = 15\,\mathrm{W\,m^{-2}\,K^{-1}}$), with the final design (Pd–Ag alloy, 20 traces, Chebyshev $p = 1.3$, $V_{\text{in}} = 12\,\mathrm{V}$). Four physics-controlled mesh levels were tested, from Extra Coarse to Extra Fine.

Table 3.1: Mesh convergence results (stationary EC + HT, Pd–Ag alloy, 20 traces, Chebyshev $p = 1.3$, $V_{\text{in}} = 12\,\mathrm{V}$, Scenario 1 conditions).

Mesh level	**Domain elements**	T_{min} **(K)**	ΔT_{min} **vs Extra Fine**
Extra Coarse	185,852	281.607	0.004 K
Coarse	241,089	281.609	0.002 K
Fine	274,402	281.611	$<$0.001 K
Extra Fine	319,050	281.611	—

The minimum surface temperature varies by only $0.004\,\mathrm{K}$ across all four mesh levels, confirming that the solution is fully converged even at Extra Coarse resolution for this geometry. The Fine mesh (274,402 elements) is selected for all simulations as it matches the Extra Fine result to within $0.001\,\mathrm{K}$ while keeping solve times practical.
It should be noted that mesh resolution is more critical for configurations where the result falls close to a threshold. For the Nichrome V material near the dew point:

- **Nichrome V, 18 traces, $12\,\mathrm{V}$:** A coarse mesh predicted a defrost time of 48 minutes, while the Fine mesh reveals that the steady-state $T_{\text{min}} = 283.09\,\mathrm{K}$ remains *below* the dew point — the window never fully defrosts.
- **Nichrome V, 16 traces, $14.5\,\mathrm{V}$:** The coarse mesh predicted "Never", while the Fine mesh gives 59 minutes ($T_{\text{min}} = 283.152\,\mathrm{K}$, barely above the dew point).

These borderline cases confirm that the Fine mesh is necessary for reliable results near the dew point threshold, and justify its use throughout this study.

3.2 Comparison to Literature

The model physics and boundary condition values are validated against published experimental and numerical studies.
The convection coefficients used in this model ($h_{\text{ext}} = 15\,\mathrm{W\,m^{-2}\,K^{-1}}$, $h_{\text{int}} = 7\,\mathrm{W\,m^{-2}\,K^{-1}}$) are compared with experimental measurements by Hożejowska et al. [4], who reported measured h values for automotive glass under various driving conditions. Their results support the order of magnitude used in this study, with exterior values in the range $10\,\mathrm{W\,m^{-2}\,K^{-1}}$ to $25\,\mathrm{W\,m^{-2}\,K^{-1}}$ depending on vehicle speed.
Rühl et al. [6] performed coupled finite element simulations of electrically heated glass panes with resistive lines, validated against infrared thermography and thermocouple measurements. Their modelling approach —

coupling electric currents with heat transfer via Joule heating in a 2D representation — is directly analogous to the physics setup used in this work.
Koshlak et al. [7] measured that 83–85% of the heat generated in electrically heated automotive glass is directed inward (toward the cabin side), which is consistent with the asymmetric convection coefficients used in this model ($h_{\text{int}} < h_{\text{ext}}$, meaning less heat loss to the interior).

3.3 Back-of-Envelope Power and Temperature Estimate

An independent analytical estimate of the steady-state average temperature provides a direct check on the numerical results. The approach is to compute the total Joule heating power from the trace resistance and compare the resulting energy balance temperature with the COMSOL output.
Each trace has electrical resistance:

$$R_{\text{trace}} = \frac{L_{\text{glass}}}{\sigma_{\text{trace}} \cdot w_{\text{trace}} \cdot t_{\text{trace}}} \tag{3.1}$$

For the Sim 1 baseline (silver paste, $\sigma = 2 \times 10^6\,\mathrm{S\,m^{-1}}$):

$$R_{\text{trace}} = \frac{1.5}{2 \times 10^6 \times 1.7 \times 10^{-3} \times 0.11 \times 10^{-3}} = 4.01\,\Omega \tag{3.2}$$

The power per trace is $P_{\text{trace}} = V_{\text{in}}^2/R_{\text{trace}} = 144/4.01 = 35.9\,\mathrm{W}$, and the total power for 12 traces is:

$$P_{\text{total}} = 12 \times 35.9 = 431\,\mathrm{W} \tag{3.3}$$

Note that this physical power is correctly reproduced in the 2D COMSOL model through the σ_{eff} correction (Equation (1.1)): the effective resistance in the model is $R_{\text{model}} = L_{\text{glass}}/(\sigma_{\text{eff}} \cdot w_{\text{trace}} \cdot t_{\text{glass}}) = L_{\text{glass}}/(\sigma_{\text{trace}} \cdot w_{\text{trace}} \cdot t_{\text{trace}}) = R_{\text{trace}}$.
At steady state, the total Joule heating must equal the convective losses from both sides of the glass:

$$P_{\text{total}} = (h_{\text{ext}} + h_{\text{int}}) \cdot A_{\text{glass}} \cdot (T_{\text{avg}} - T_{\text{weighted}}) \tag{3.4}$$

where $A_{\text{glass}} = L_{\text{glass}} \times H_{\text{glass}} = 1.2\,\mathrm{m}^2$ and the convection-weighted ambient temperature is:

$$T_{\text{weighted}} = \frac{h_{\text{ext}} \cdot T_{\text{amb}} + h_{\text{int}} \cdot T_{\text{cabin}}}{h_{\text{ext}} + h_{\text{int}}} = \frac{15 \times 275.15 + 7 \times 293.15}{22} = 280.9\,\mathrm{K} \tag{3.5}$$

Solving for T_{avg}:

$$T_{\text{avg}} = T_{\text{weighted}} + \frac{P_{\text{total}}}{(h_{\text{ext}} + h_{\text{int}}) \cdot A_{\text{glass}}} = 280.9 + \frac{431}{22 \times 1.2} = 280.9 + 16.3 = 297.2\,\mathrm{K} \tag{3.6}$$

The COMSOL surface-average temperature for Sim 1 (12 traces, silver paste, $12\,\mathrm{V}$) is $T_{\text{avg}} = 297.13\,\mathrm{K}$. The back-of-envelope estimate gives $297.2\,\mathrm{K}$ — a deviation of only $0.07\,\mathrm{K}$ ($< 0.03\%$). This excellent agreement confirms that:

- The σ_{eff} correction correctly maps the 3D trace geometry to the 2D model
- The Joule heating and convective boundary conditions are consistently implemented
- The total energy balance of the model is accurate

Table 3.2: Back-of-envelope validation of the average temperature for selected configurations.

Material	n_{traces}	P_{total} (W)	$T_{\text{avg}}^{\text{analytical}}$ (K)	$T_{\text{avg}}^{\text{COMSOL}}$ (K)	**Error**
Silver paste	12	431	297.2	297.13	0.03%
Pd–Ag alloy	14	578	302.8	*— extract from COMSOL*	—

STILL DO: EXTRACT T_{avg} FROM COMSOL FOR Pd–Ag 14-TRACE AND 20-TRACE CONFIGURATIONS TO COMPLETE THE TABLE

3.4 Sensitivity Analysis

The parametric sweeps performed in Sim 2 and Sim 2b serve as a de facto sensitivity analysis, showing how the minimum surface temperature responds to changes in the three main design variables.

Table 3.3: Sensitivity of minimum surface temperature to design parameters (Pd–Ag alloy, baseline: 14 traces, $12\,\mathrm{V}$).

Parameter change	ΔT_{min} **(K)**	**Sensitivity**
n_{traces}: 14 $\rightarrow$ 16 (+2 traces)	+1.40	$0.70\,\mathrm{K}$/trace
V_{in}: 12 $\rightarrow$ $14.5\,\mathrm{V}$ (+$2.5\,\mathrm{V}$)	+1.16	$0.46\,\mathrm{K\,V^{-1}}$
Material: Pd–Ag $\rightarrow$ Nichrome V	-1.49	—

The results confirm that **the number of traces is the most influential parameter**, followed by applied voltage. The choice of trace material has a strong effect through its conductivity: higher conductivity produces more Joule heating per unit length of trace.
The external heat transfer coefficient h_{ext} is also a sensitive parameter (as noted in the literature [4]), since it directly controls the rate of heat loss from the glass. This was systematically addressed in Sim 6 (Section 2.8), which swept h_{ext} alongside T_{amb} and T_{dew} across five realistic weather scenarios, confirming that the final design remains robust across all tested conditions.

Chapter 4

Results and Discussion

4.1 Transient Defrost Performance

The transient defrost times for all tested configurations are presented in Tables 4.1 and 4.2. These results are from the fine-mesh simulation (189,192 domain elements).

Table 4.1: Defrost time (minutes) for Pd–Ag alloy ($\sigma = 2.3 \times 10^6\,\mathrm{S\,m^{-1}}$). Defrost is defined as the time for T_{min} to reach $283.15\,\mathrm{K}$.

n_{traces}	$12\,\mathrm{V}$	$14.5\,\mathrm{V}$
14	41	22
16	22	16
18	16	12
20	**12**	10
22	10	**8**

Table 4.2: Defrost time (minutes) for Nichrome V ($\sigma = 9.3 \times 10^5\,\mathrm{S\,m^{-1}}$). “Never” indicates that T_{min} does not reach $283.15\,\mathrm{K}$ within $60\,\mathrm{min}$.

n_{traces}	$12\,\mathrm{V}$	$14.5\,\mathrm{V}$
14	Never	Never
16	Never	59
18	Never	25
20	28	18
22	20	14

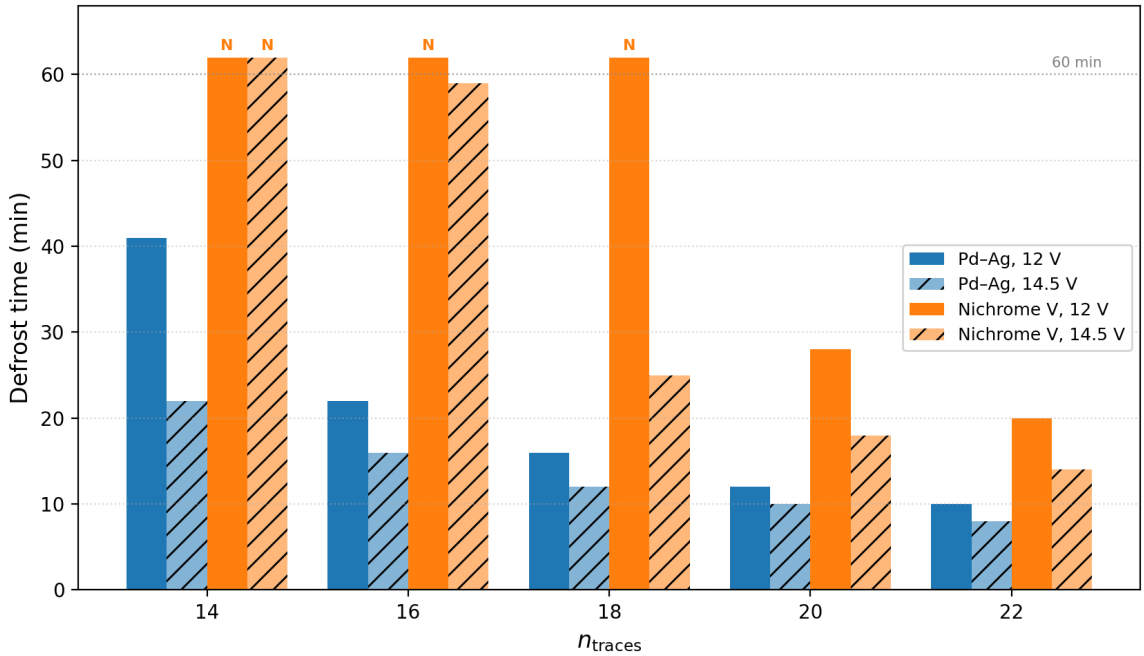


Figure 4.1: Defrost time comparison between Pd–Ag alloy and Nichrome V at $12\,\mathrm{V}$ and $14.5\,\mathrm{V}$. "N" indicates the configuration never reaches the dew point within $60\,\mathrm{min}$.

Pd–Ag alloy consistently defrosts faster than Nichrome V at the same trace count and voltage. Nichrome V requires ≥ 20 traces at $12\,\mathrm{V}$ to defrost at all within 60 minutes, while Pd–Ag defrosts with as few as 14 traces (in 41 minutes).

4.2 Temperature Distribution

Figure 4.2 shows the steady-state temperature distribution for the recommended design (Pd–Ag alloy, 20 traces, $12\,\mathrm{V}$).

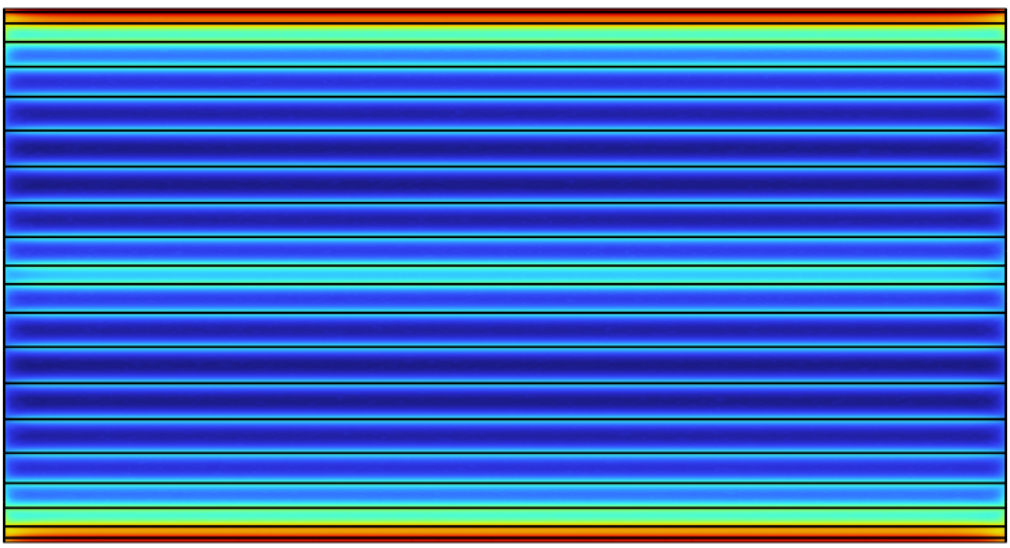


Figure 4.2: Steady-state temperature distribution for the recommended design: Pd–Ag alloy, 20 traces, $V_{\text{in}} = 12\,\mathrm{V}$. The minimum temperature ($288.2\,\mathrm{K}$) is well above the dew point ($283.15\,\mathrm{K}$).

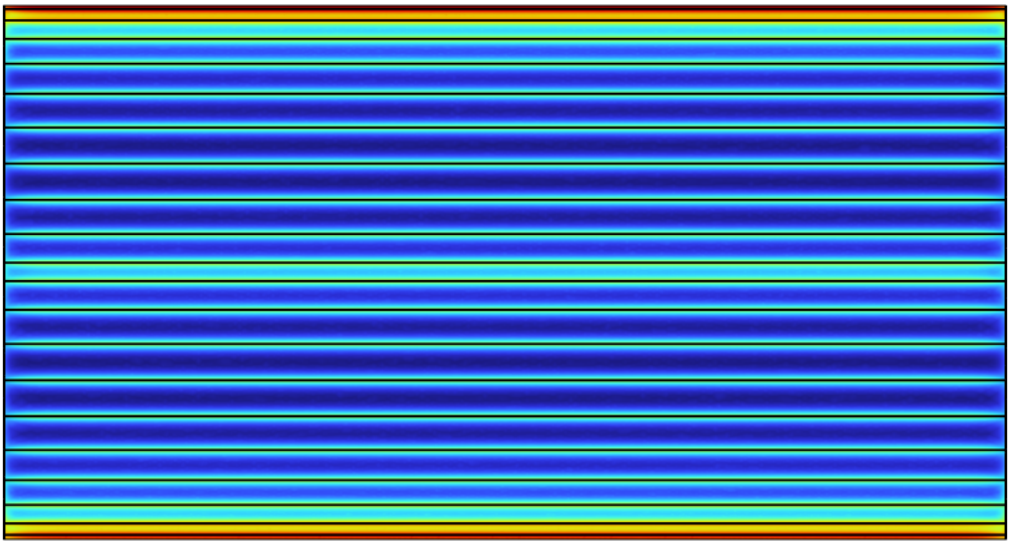


Figure 4.3: Temperature distribution at $t = 12\,\mathrm{min}$ (moment of defrost) for Pd–Ag alloy, 20 traces, $V_{\text{in}} = 12\,\mathrm{V}$.

4.3 Effect of Trace Material Conductivity

A key finding of this study is the strong dependence of defroster performance on the trace electrical conductivity. The Joule heating power per unit length of trace scales as:

$$P \propto \frac{V_{\text{in}}^2 \cdot \sigma_{\text{eff}}}{L_{\text{trace}}} \tag{4.1}$$

where σ_{eff} is the effective 2D conductivity from Equation (1.1). This explains why:

- Materials with very high conductivity (Silver: 5.75×10^7, Gold: 4.2×10^7 S/m) generate excessive heating, reaching temperatures of 415–465 K at 20 traces — far above safe operating limits.
- The optimal conductivity range for this geometry and voltage is $\sim 10^5$–10^7 S/m, which corresponds to thick-film paste materials such as Pd–Ag alloy and Nichrome V.

Figure 4.4 shows the simulated steady-state T_{min} for all four EduPack materials at $n_{\text{traces}} = 20$, $V_{\text{in}} = 12\,\mathrm{V}$, plotted against their electrical conductivity on a logarithmic scale. The two viable materials cluster just above the dew point line, while Silver and Gold overheat catastrophically.

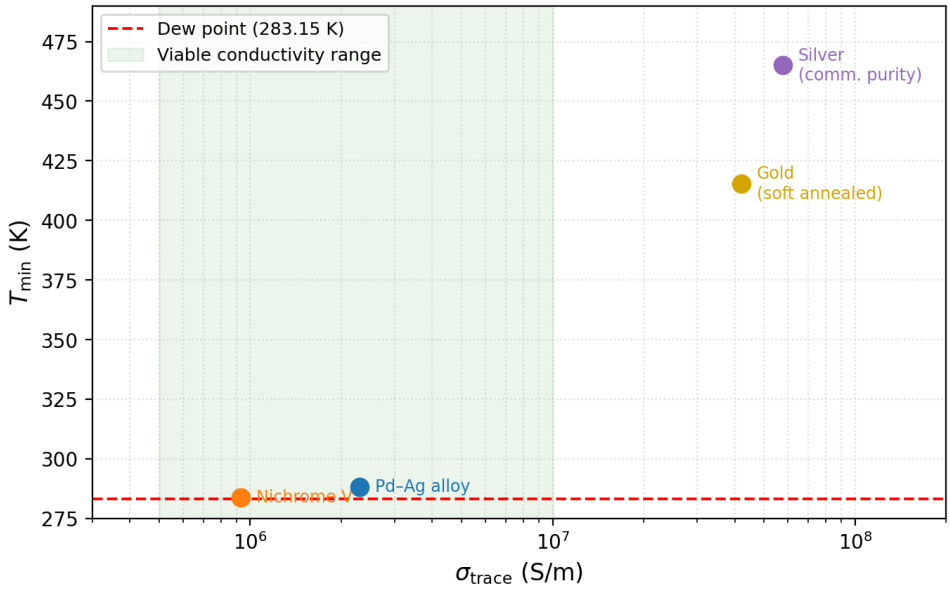


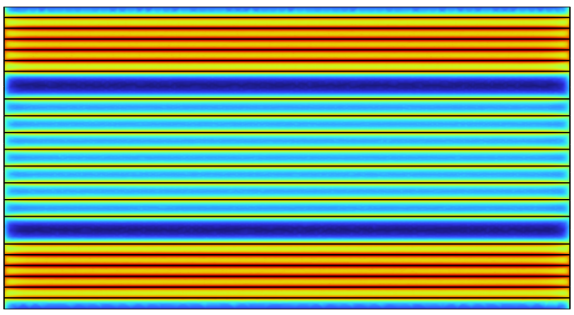
Figure 4.4: Steady-state T_{min} vs. trace electrical conductivity for all four EduPack materials ($n_{\text{traces}} = 20$, $V_{\text{in}} = 12\,\text{V}$). The green shaded region indicates the viable conductivity range. Silver and Gold fall far above the safe operating temperature.

4.4 Trace Pattern Optimisation

Two approaches to non-uniform trace spacing were investigated: a discrete zoned layout (Approach A, Sim 4) and a continuous Chebyshev distribution (Approach B, Sim 5). Both use 20 traces of Pd–Ag alloy at $V_{\text{in}} = 12\,\text{V}$ to allow direct comparison with the uniform baseline.

The zoned configuration places 6 traces in the top $200\,\text{mm}$, 8 traces in the middle $400\,\text{mm}$, and 6 traces in the bottom $200\,\text{mm}$. This reduces the edge gap from $38.1\,\text{mm}$ (uniform) to $28.6\,\text{mm}$ while widening the centre gap to $44.4\,\text{mm}$.

The time-dependent simulation shows that this configuration defrosts in $11\,\text{min}$ (vs. $12\,\text{min}$ for uniform) and reaches a steady-state T_{min} of $289.3\,\text{K}$ (vs. $288.2\,\text{K}$). The improvement is modest: $1\,\text{min}$ faster defrost and $1.1\,\text{K}$ higher T_{min}.



TODO (Sebtain): Sim3 uniform 20 traces at defrost moment ($t = 12\,\text{min}$)

Figure 4.5: Temperature distribution at the moment of defrost. (a) Zoned 6/8/6 at $t = 11\,\text{min}$. (b) Uniform baseline at $t = 12\,\text{min}$. Both: Pd–Ag alloy, 20 traces, $12\,\text{V}$.

The Chebyshev approach (Equation (2.1)) is far more effective. With the optimal exponent $p = 1.3$, the edge gaps shrink to $\sim 3\,\text{mm}$ while the widest centre gap reaches $\sim 55\,\text{mm}$. Despite this extreme non-uniformity, the

glass conductivity bridges the centre gap and the overall T_{min} increases substantially because the previously cold edges are now directly heated.

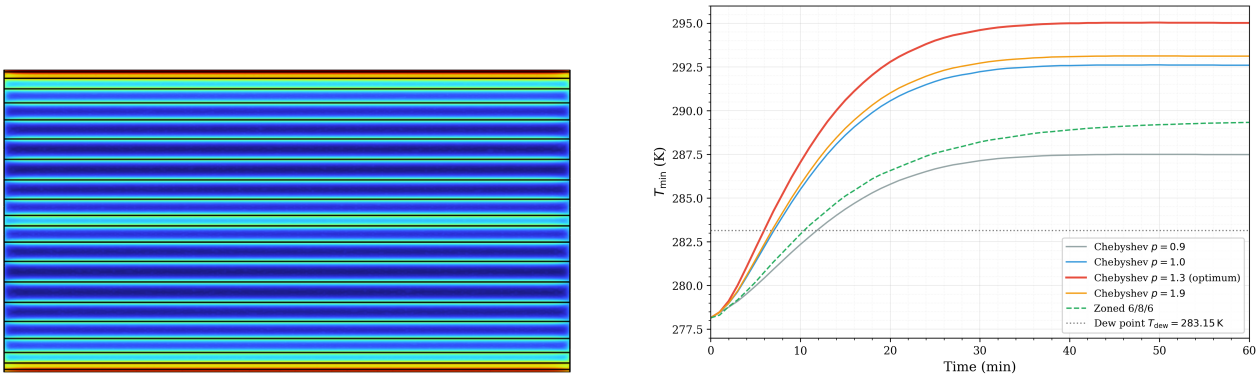


Figure 4.6: Chebyshev spacing results. (a) Temperature distribution at $t = 6\,\text{min}$ for $p = 1.3$. (b) Transient T_{min} evolution for selected p values; the dashed line is the dew point.

Table 4.3 summarises the performance of all trace patterns.

Table 4.3: Comparison of trace spacing approaches (Pd–Ag alloy, 20 traces, $V_{\text{in}} = 12\,\text{V}$).

Approach	**Defrost time**	**Steady-state T_{min} (K)**	**Improvement**
Uniform (baseline)	12 min	288.2	—
Zoned A2 (6/8/6)	11 min	289.3	-8% time, $+1.1\,\text{K}$
Chebyshev $p = 1.0$	8 min	292.6	-33% time, $+4.4\,\text{K}$
Chebyshev $p = 1.3$	**6 min**	**295.0**	**-50% time**, $+6.8\,\text{K}$

The Chebyshev $p = 1.3$ spacing halves the defrost time compared to uniform spacing while raising T_{min} by $6.8\,\text{K}$. This is achieved without any change in material, voltage, or number of traces — purely through optimised trace placement. The result demonstrates that **geometry optimisation** using COMSOL's parametric capabilities can be as impactful as material or electrical optimisation.

4.5 Discussion

4.5.1 Strengths

- With Chebyshev spacing ($p = 1.3$), the final design (Pd–Ag alloy, 20 traces, $12\,\text{V}$) achieves a steady-state T_{min} of $295.0\,\text{K}$ — $12\,\text{K}$ above the dew point — and defrosts in only $6\,\text{min}$. With uniform spacing, the same configuration takes $12\,\text{min}$. The improvement comes purely from trace placement, at zero additional cost.
- The thermal stresses are within safe limits (safety factor $\sim 3.1\times$) and the design operates at nominal $12\,\text{V}$ battery voltage, meaning it works even with the engine off.
- The parametric approach (sweeping material, trace count, voltage, and spacing exponent) provides a comprehensive design map that allows easy trade-off analysis for different requirements.
- The multiphysics coupling between Electric Currents, Heat Transfer, and Solid Mechanics captures the essential physics of the defroster operation.
- The mesh convergence study confirms that results are mesh-independent for the recommended configurations.
- The boundary conditions (h_{ext}, h_{int}) are supported by published experimental data [4].
- **The design is robust across real-world weather conditions (Sim 6).** All five scenarios — spanning ambient temperatures from $-20\,°\text{C}$ to $10\,°\text{C}$, external convection coefficients from $10\,\text{W}\,\text{m}^{-2}\,\text{K}^{-1}$ to $40\,\text{W}\,\text{m}^{-2}\,\text{K}^{-1}$, and physically realistic dew points — were defrosted within $8\,\text{min}$, with a minimum steady-state margin of $5.4\,\text{K}$ above the dew point (freezing rain scenario).
- The sequential solve strategy for the stress analysis overcomes the COMSOL 6.2 spatial frame limitation while still capturing the thermal stress state.

4.5.2 Limitations, Future Work, and Broader Implications

- **2D approximation:** The model uses a flat 2D geometry. A real rear window is curved, which affects heat flux distribution and stress concentrations. A 3D simulation with curved glass would be more realistic.
- **Fixed boundary conditions:** The fixed constraint at the glass edges overestimates the peak stress. Modelling the rubber seal as an elastic support would give more realistic stress values.
- **Constant heat transfer coefficients:** In reality, h_{ext} depends on vehicle speed (forced convection) and h_{int} depends on cabin HVAC settings. Sim 6 addressed the former by sweeping h_{ext} from $10\,\mathrm{W\,m^{-2}\,K^{-1}}$ to $40\,\mathrm{W\,m^{-2}\,K^{-1}}$ across five weather scenarios; the effect of varying h_{int} with HVAC state remains as future work.
- **Material property validation:** The EduPack values for Pd–Ag alloy are for bulk material. Screen-printed thick-film pastes may have different effective properties. Experimental validation with a physical prototype is recommended.
- **Contact resistance:** The model does not account for electrical contact resistance between traces and bus bars, which may reduce the effective heating power.

The Chebyshev spacing result has broader implications beyond this specific defroster. The key insight is that **non-uniform spatial distribution of heat sources**, optimised using a single continuous parameter, can be as effective as changing materials or operating conditions. This principle applies to any distributed heating or cooling problem where temperature uniformity is the objective.
In the context of automotive defrosters, the Chebyshev approach could be directly applied to front windshield defrosters (where driver visibility constraints are even more stringent) or to heated mirrors. The p-parameter provides a simple manufacturing specification: a single number that fully defines the trace layout for any given window height and trace count.
Furthermore, the observation that bulk-conductivity materials (Silver, Gold) overheat catastrophically in this geometry has a practical consequence: screen-printed thick-film pastes are not merely a cost-saving alternative to bulk metals — they are *functionally necessary* for the correct thermal behaviour. This finding is consistent with the industrial practice of using Pd–Ag and Ag paste formulations with controlled conductivity [8].
STILL DO: ELEMENT ORDER CONVERGENCE ANALYSIS — comparing linear vs. quadratic elements for the Joule heating/temperature solution to verify that the default element order is sufficient.
STILL DO: ADAPTIVE MESH REFINEMENT ANALYSIS — using COMSOL's adaptive mesh refinement to verify error control near the trace boundaries where temperature gradients are steepest.

Chapter 5

Conclusion

This report presents the design and optimisation of a resistive rear window defroster using coupled multi-physics finite element modelling in COMSOL Multiphysics 6.2. The design was developed iteratively through six simulations addressing thermal performance, material selection, structural integrity, transient response, trace pattern optimisation, and weather condition robustness.
The key findings of this study are:

1. **The number of heating traces is the most influential design parameter.** Increasing the trace count from 12 to 20 raises T_{min} by over $6\,\mathrm{K}$ at $12\,\mathrm{V}$, whereas increasing voltage from 12 to $14.5\,\mathrm{V}$ at 14 traces yields only $1.2\,\mathrm{K}$.

2. **Pd–Ag alloy ($\sigma = 2.3 \times 10^6\,\mathrm{S\,m^{-1}}$) is the optimal trace material.** It reaches the dew point with 14 traces at $12\,\mathrm{V}$, while Nichrome V requires 18–20 traces. Bulk conductivity materials (Silver, Gold) overheat catastrophically and are not viable for this geometry.

3. **Both shortlisted materials pass the stress analysis** with safety factors exceeding $3\times$, confirming that thermal expansion mismatch between the traces and glass does not pose a structural risk.

4. **The recommended design** is:
   - Material: Pd–Ag alloy
   - Number of traces: 20
   - Voltage: $12\,\mathrm{V}$ (nominal battery)
   - Steady-state T_{min}: $288.2\,\mathrm{K}$ ($5\,\mathrm{K}$ above dew point)
   - Defrost time: 12 minutes
   - Stress safety factor: $\sim 3.1\times$

5. **Alternative high-performance design:** Pd–Ag alloy, 22 traces, $14.5\,\mathrm{V}$ — achieves 8-minute defrost time when the engine is running.

6. **The final design is robust across all tested weather conditions (Sim 6).** Five scenarios ranging from mild autumn ($10\,^\circ\mathrm{C}$) to extreme cold ($-20\,^\circ\mathrm{C}$) and freezing rain ($h_{\text{ext}} = 40\,\mathrm{W\,m^{-2}\,K^{-1}}$) were tested, each with a physically realistic dew point [5]. All five scenarios defrost within $8\,\mathrm{min}$ with a minimum steady-state margin of $5.4\,\mathrm{K}$ above the dew point, confirming that the design is not tuned to a single favourable condition.

7. **Chebyshev trace spacing halves the defrost time.** The non-uniform Chebyshev spacing (Equation (2.1)) with exponent $p = 1.3$ reduces the defrost time from $12\,\mathrm{min}$ to $6\,\mathrm{min}$ and raises T_{min} by $6.8\,\mathrm{K}$ — without changing the number of traces, material, or voltage. The discrete zoned approach (6/8/6) provides only a modest $1\,\mathrm{min}$ improvement, confirming that a continuous optimisation is needed. The optimal exponent ($p = 1.3$) was found via a 16-point parametric sweep, exploiting COMSOL's ability to parametrically update the geometry.

8. **Final recommended design:**
   - Material: Pd–Ag alloy ($\sigma = 2.3 \times 10^6\,\mathrm{S\,m^{-1}}$)
   - Number of traces: 20
   - Spacing: Chebyshev with $p = 1.3$

- Voltage: $12\,\mathrm{V}$ (nominal battery)
- Steady-state T_{min}: $295.0\,\mathrm{K}$ ($12\,\mathrm{K}$ above dew point)
- Defrost time: $6\,\mathrm{min}$
- Stress safety factor: $\sim 3.1\times$
- Window obscuration: 4.25% ($< 5\%$ SAE J216 limit)

The simulation methodology — coupling Electric Currents, Heat Transfer, and Solid Mechanics with parametric sweeps over material properties, geometry, and operating conditions — provides a systematic framework for defroster design. The model was validated through mesh convergence, back-of-envelope analytical estimates (agreeing to within 0.03%), and comparison with published experimental data.
The key insight of this study is that **geometry optimisation** (trace placement) can be equally or more impactful than material or electrical optimisation. The Chebyshev spacing approach provides a mathematically principled, single-parameter design specification that can be applied to any rectangular heated-glass application. Future work should extend the model to 3D curved glass, validate against physical prototypes, and investigate the effect of varying interior convection (h_{int}) with different cabin HVAC states.

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